Sequences and Series

ANSWER KEY

Sequence limits

1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$ Converges to 3 (divide by n).

b $a_n = \frac{n^2}{2^n}$ Converges to 0 (exponential growth dominates polynomial).

c $a_n = (-1)^n$ Diverges (oscillates between -1 and 1).

2 Evaluate $\lim_{n\to\infty}\left(1+\frac{3}{n}\right)^n$.

This is the standard limit form giving e^3.

Series and partial sums

3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

Diverges — this is the harmonic series, a standard example of a divergent series with terms going to 0.

4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

$|r| = \frac{2}{5} < 1$: $S = \frac{3}{1-2/5} = 5$.

5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The partial sums telescope: $S_N = 1 - \frac{1}{N+1} \to 1$.

Monotonic and bounded sequences

6 Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n.

$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$, so $a_{n+1} > a_n$: the sequence is increasing.

7 Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.

By the Monotonic Convergence Theorem, a bounded, monotonic sequence converges. $\lim_{n\to\infty}\frac{n}{n+1}=1$ (divide numerator and denominator by n).

More series practice

8 Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

$\sum\left(\frac{1}{2}\right)^n = \frac{1/2}{1-1/2} = 1$. $\sum\left(\frac{1}{3}\right)^n = \frac{1/3}{1-1/3} = \frac{1}{2}$. Total $= 1 + \frac{1}{2} = \frac{3}{2}$.

9 A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.

Total $= 10 + 2\left(10(0.6) + 10(0.6)^2 + \cdots\right) = 10 + 2 \cdot \frac{6}{1 - 0.6} = 10 + 2(15) = 40$ m.

Recursive sequences

10 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2, a_3, and a_4.

$a_2 = \frac{2+6}{2} = 4$. $a_3 = \frac{4+6}{2} = 5$. $a_4 = \frac{5+6}{2} = 5.5$.

11 Assuming the recursive sequence above converges to a limit L, find L by setting $L = \frac{L+6}{2}$ and solving.

$2L = L + 6$, so $L = 6$.

Repeating decimals as series

12 Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.

$S = \frac{45/100}{1 - 1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$.

Visualizing partial sums of a series

13 The graph shows the first six partial sums $S_n = \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.

The partial sums appear to approach 1. Using the geometric series formula: $S = \frac{1/2}{1 - 1/2} = 1$, confirming the visual pattern.

Series with factorial and combined terms

14 Evaluate $\sum_{}^{\infty} \frac{1}{2^n} - \sum_{}^{\infty} \frac{1}{3^n}$ by evaluating each geometric series separately.

$\sum \frac{1}{2^n} = \frac{1}{1 - 1/2} = 2$. $\sum \frac{1}{3^n} = \frac{1}{1 - 1/3} = \frac{3}{2}$. Difference $= 2 - \frac{3}{2} = \frac{1}{2}$.

15 A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n \to \infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.

$a_n = 2 + \frac{(-1)^n}{n}$. As $n \to \infty$, $\frac{(-1)^n}{n} \to 0$ (bounded numerator divided by an unbounded denominator), so $\lim_{n \to \infty} a_n = 2$.

The Squeeze Theorem for sequences

16 Use the squeeze theorem to find $\lim_{n\to\infty}\frac{\sin n}{n}$, using the fact that $-1 \le \sin n \le 1$ for all n.

$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$ for $n \ge 1$. Since $\lim_{n\to\infty}\left(\pm\frac{1}{n}\right)=0$, by the squeeze theorem $\lim_{n\to\infty}\frac{\sin n}{n} = 0$.

17 Use the squeeze theorem to find $\lim_{n\to\infty}\frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \le \frac{1}{n}$ for $n \ge 1$.

Since $0 \le \frac{n!}{n^n} \le \frac{1}{n}$ and $\lim_{n\to\infty}\frac{1}{n}=0$, by the squeeze theorem $\lim_{n\to\infty}\frac{n!}{n^n} = 0$.

Applications of infinite series

18 A patient takes a 100 mg dose of medication every 8 hours. Between doses, 75% of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).

Immediately after the nth dose, the amount is $100 + 100r + 100r^2 + \cdots = \frac{100}{1 - 0.25} = \frac{100}{0.75} \approx 133.33$ mg in the long run.